Lecture 4:
Circular Motion
Conical Pendulum
Banked Curve
related Problem Solving

Circular Motion

Acoording to Newton's second law in its vector form $\sum \vec{F} = m\vec{a}$ the acceleration and the net force must have same direction.

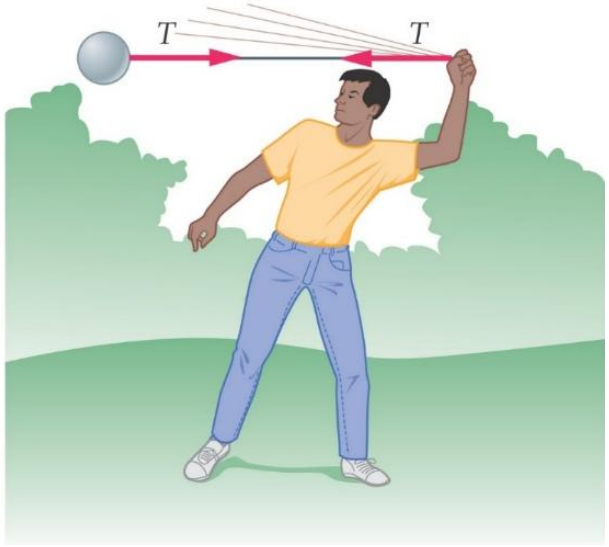
In case of circular motion at constant speed the net force must points towards the center of the circle.

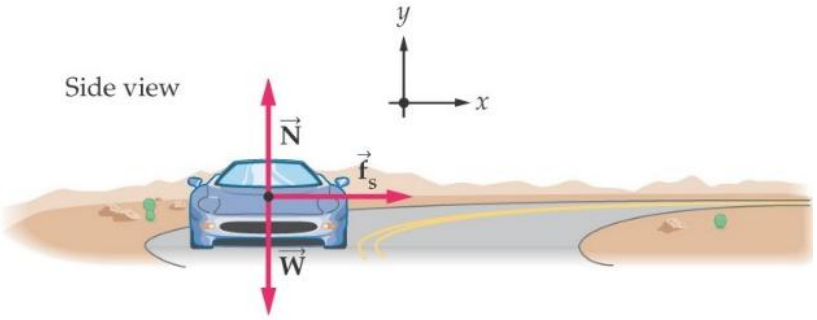
$$f_{\text{cp}} = ma_{\text{cp}} = m\frac{v^2}{r}$$

f_{cp} is called centripetal force.

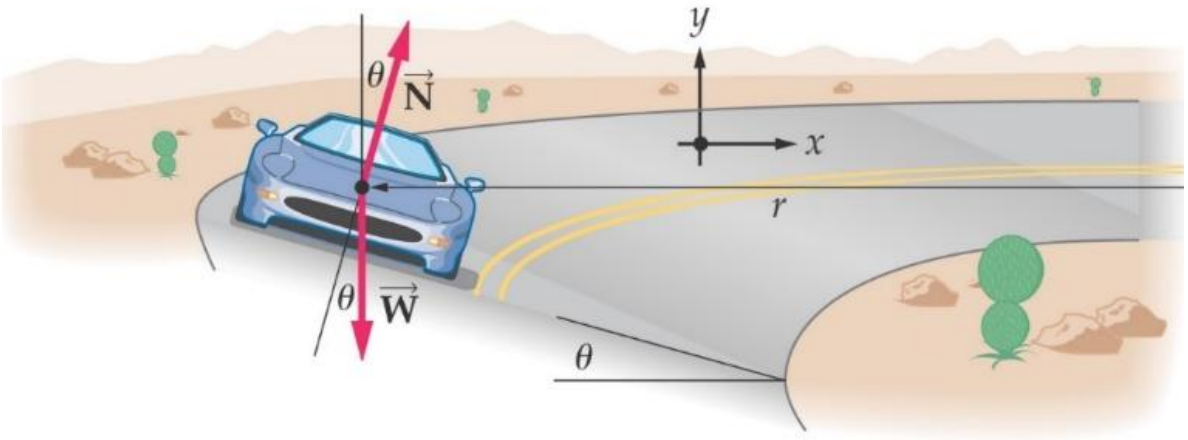
Circular Motion

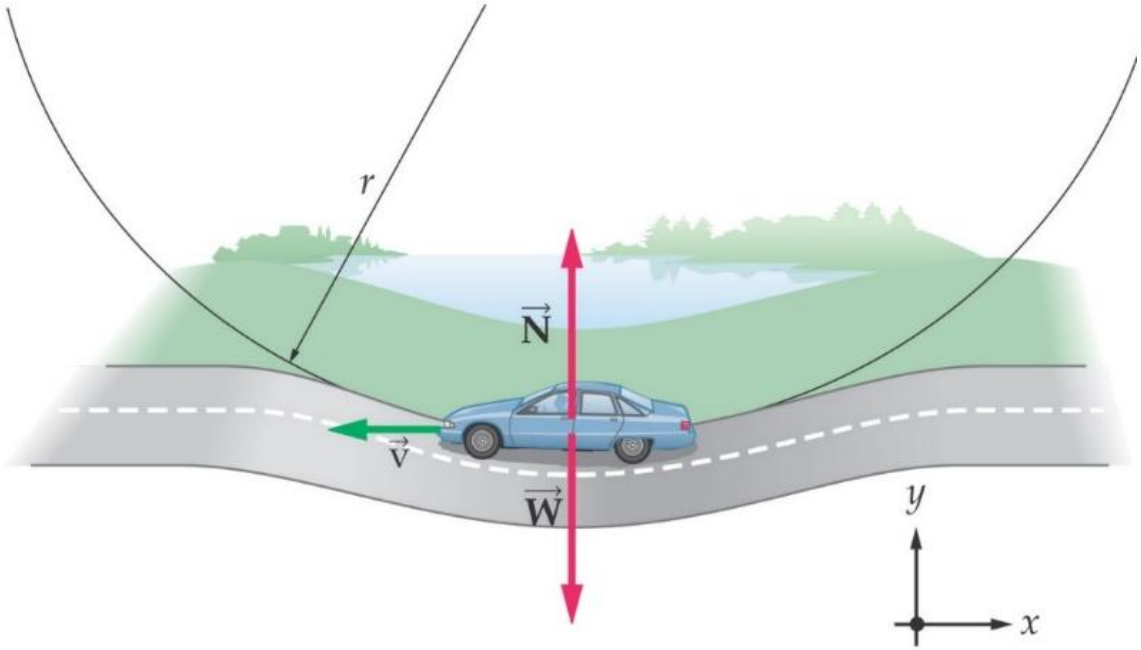
This force may be provided by the tension in a string, the normal force, or friction, among others.





Circular Motion



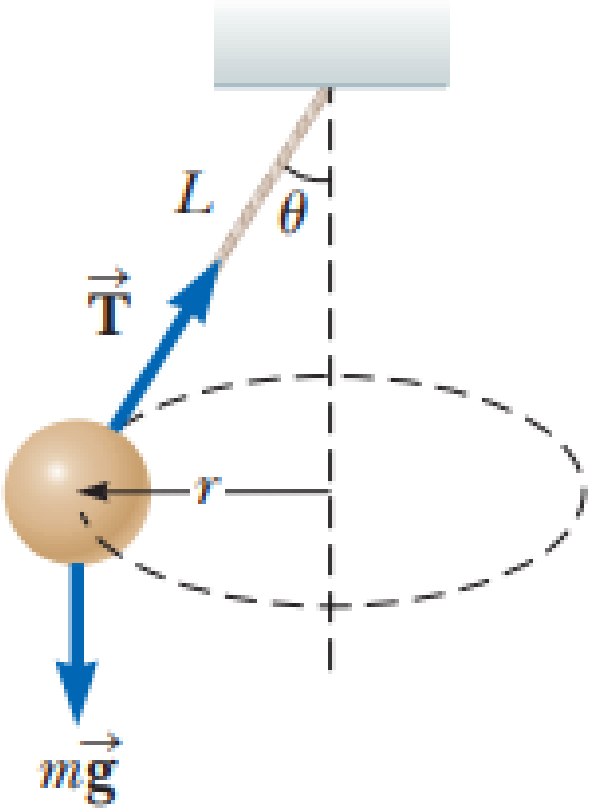


THE DYNAMICS OF UNIFORM CIRCULAR MOTION

Conical Pendulum

Figure shows a small body of mass m moving in a horizontal circle with constant speed v at the end of a string of length L. As the body swings around, the string sweeps over the surface of imaginary cone. This device is called conical pendulum.

To calculate time required to complete one revolution of the body if the strings makes angle θ with the vertical the radius of the circular path is $R = Lsin\theta$.



THE DYNAMICS OF UNIFORM CIRCULAR MOTION

Conical Pendulum

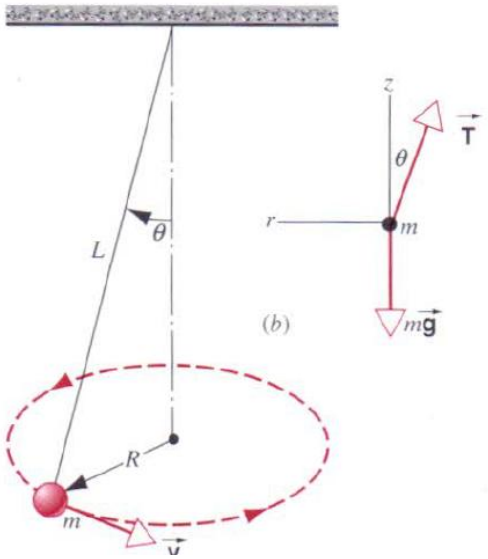
$$\sum F_r = ma_r$$

$$-Tsin\theta = -m\frac{v^2}{R}$$

$$v^2 = \frac{Tsin\theta R}{m}$$

$$v^2 = \frac{\text{mgsin}\theta\text{R}}{mcos\theta} = gRtan\theta$$

$$v = \sqrt{gRtan\theta}$$



$$\sum F_z = ma_z$$

$$Tcos\theta - \text{mg} = 0$$

$$Tcos\theta = mg$$

$$T = \frac{mg}{cos\theta}$$

A disk of mass m on a frictionless table is attached to a hanging cylinder of mass m by a cord through a hole in the table. Find the speed with which the disk must move in a circle of radius r for the cylinder to stay at rest.

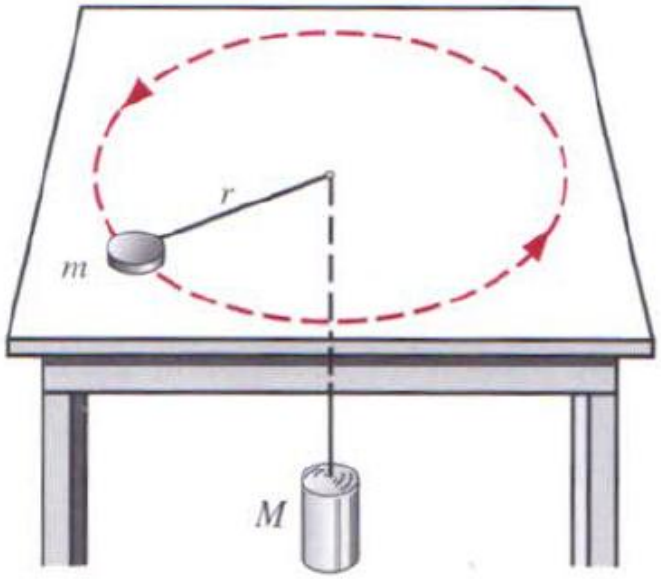


FIGURE 5-42. Exercise 39.

E5-39 There are two forces on the hanging cylinder: the force of the cord pulling up T and the force of gravity $W = Mg$. The cylinder is at rest, so these two forces must balance, or $T = W$. There are three forces on the disk, but only the force of the cord on the disk T is relevant here, since there is no friction or vertical motion.

The disk undergoes circular motion, so $T = mv^2/r$. We want to solve this for v and then express the answer in terms of m, M, r, and G.

$$v = \sqrt{\frac{Tr}{m}} = \sqrt{\frac{Mgr}{m}}.$$

The Banked Curve

Let the block in Fig. 5-20*a* represent an automobile or railway car moving at constant speed v on a *level* roadbed around a curve having a radius of curvature R. In addition to two vertical forces—namely, the weight $m\vec{\mathbf{g}}$ and a normal force $\vec{\mathbf{N}}$—a horizontal force $\vec{\mathbf{P}}$ must act on the car. The force $\vec{\mathbf{P}}$ provides the centripetal force necessary for motion in a circle. In the case of the automobile this force is supplied by a sidewise frictional force exerted by the road on the tires; in the case of the railway car the force is supplied by the rails exerting a sidewise force on the inner rims of the car's wheels. Neither of these sidewise forces can safely be relied on to be large enough at all times, and both cause

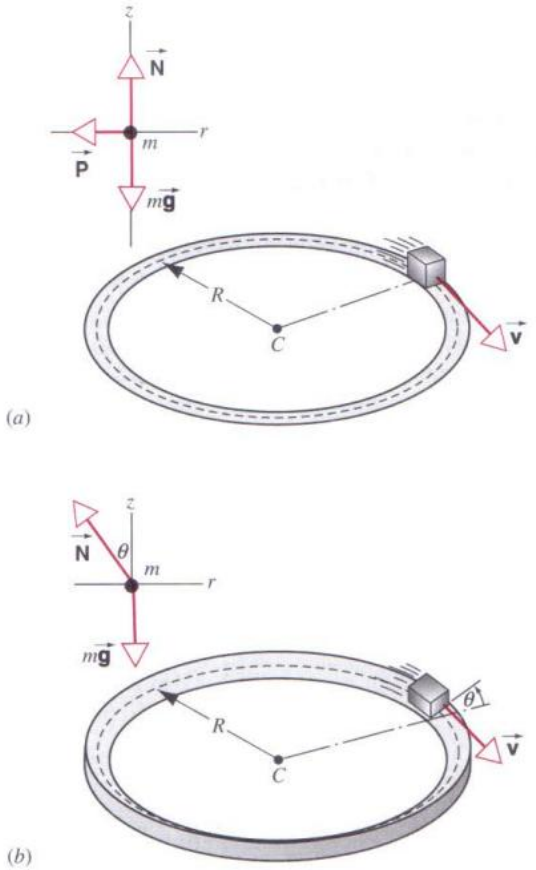


FIGURE 5-20. (*a*) A level roadbed. A free-body diagram of the moving body is shown at left. The centripetal force must be supplied by friction between tires and road. (*b*) A banked roadbed. No friction is necessary to round the curve safely.

The correct angle θ of banking in the absence of friction can be obtained as follows. We begin, as usual, with Newton's second law, and we refer to the free-body diagram shown in Fig. 5-20*b*. The radial and vertical components of the net force on the moving body are $\Sigma F_r = -N \sin \theta$ and $\Sigma F_z = N \cos \theta - mg$. As before, the radial acceleration is $a_r = -v^2/R$ and the vertical acceleration is $a_z = 0$, so we can write the components of Newton's second law as

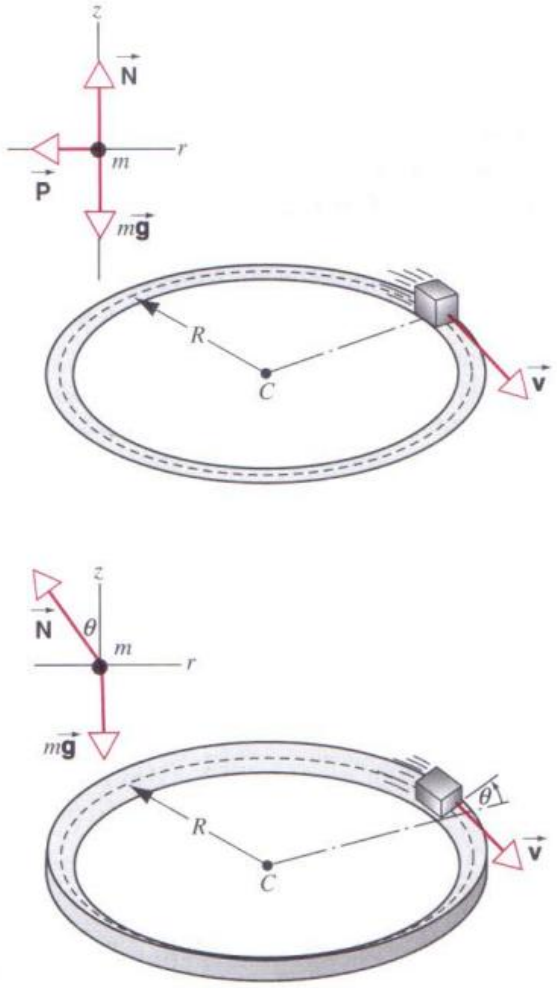
$$-N \sin \theta = ma_r = m\left(\frac{-v^2}{R}\right)$$

and

$$N \cos \theta - mg = ma_z = 0.$$

Solving these two equations for $\sin \theta$ and $\cos \theta$ and dividing the resulting expressions, we obtain

$$\tan \theta = \frac{v^2}{Rg}. \qquad (5\text{-}17)$$

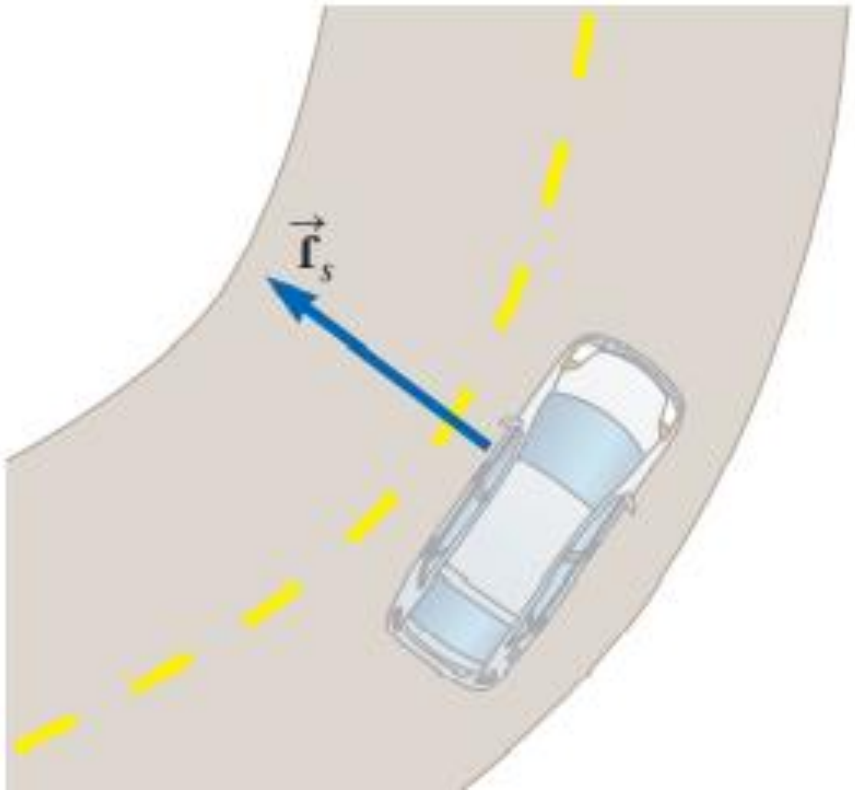


(*b*)

FIGURE 5-20. (*a*) A level roadbed. A free-body diagram of the moving body is shown at left. The centripetal force must be supplied by friction between tires and road. (*b*) A banked roadbed. No friction is necessary to round the curve safely.

Example 6.3 What Is the Maximum Speed of the Car? AM

A 1 500-kg car moving on a flat, horizontal road negotiates a curve as shown in Figure 6.4a. If the radius of the curve is 35.0 m and the coefficient of static friction between the tires and dry pavement is 0.523, find the maximum speed the car can have and still make the turn successfully.



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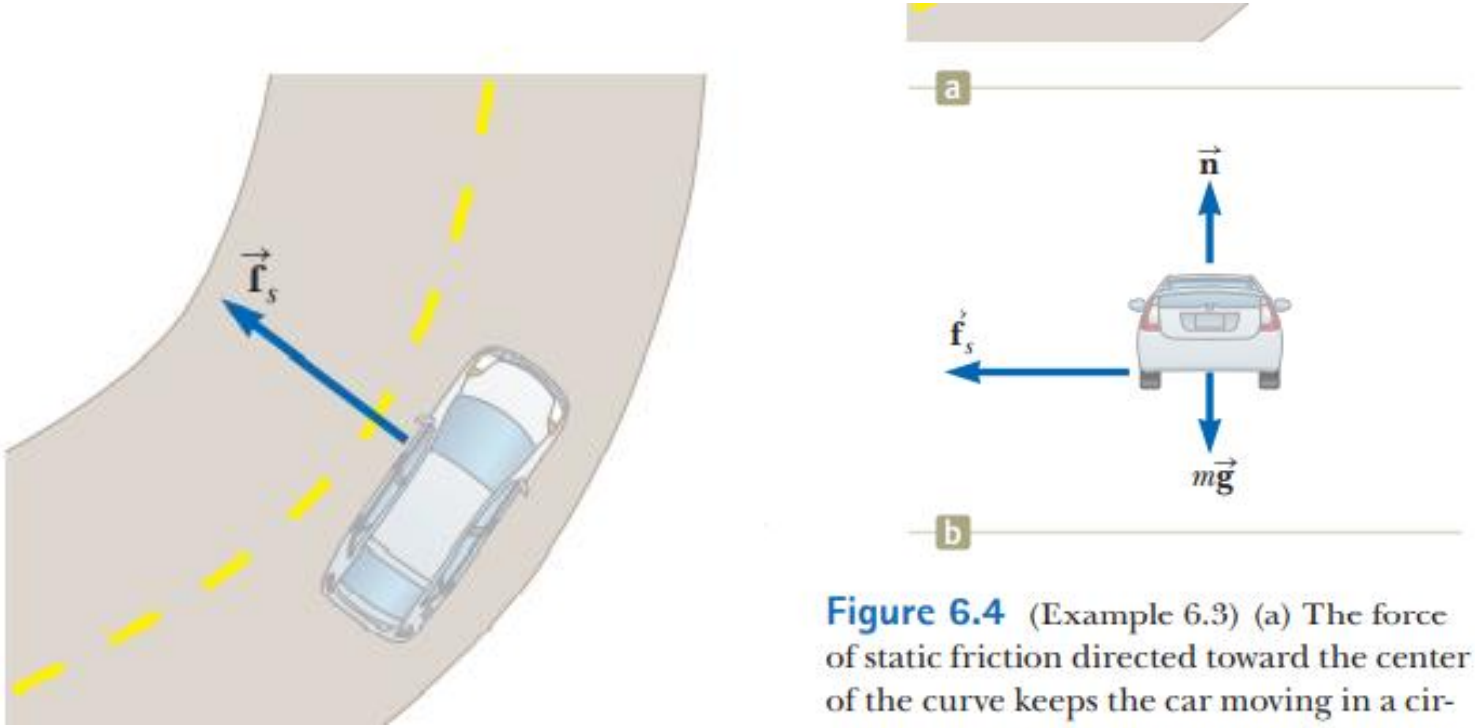


Figure 6.4 (Example 6.3) (a) The force of static friction directed toward the center of the curve keeps the car moving in a circular path. (b) The forces acting on the car.

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Apply Equation 6.1 from the particle in uniform circular motion model in the radial direction for the maximum speed condition:

$$(1)\quad f_{s,\max} = \mu_s n = m\frac{v_{\max}^2}{r}$$

Apply the particle in equilibrium model to the car in the vertical direction:

$$\sum F_y = 0 \rightarrow n - mg = 0 \rightarrow n = mg$$

Solve Equation (1) for the maximum speed and substitute for n:

$$(2)\quad v_{\max} = \sqrt{\frac{\mu_s n r}{m}} = \sqrt{\frac{\mu_s m g r}{m}} = \sqrt{\mu_s g r}$$

Substitute numerical values:

$$v_{\max} = \sqrt{(0.523)(9.80\ \text{m/s}^2)(35.0\ \text{m})} = 13.4\ \text{m/s}$$

Finalize This speed is equivalent to 30.0 mi/h. Therefore, if the speed limit on this roadway is higher than 30 mi/h, this roadway could benefit greatly from some banking, as in the next example! Notice that the maximum speed does not depend on the mass of the car, which is why curved highways do not need multiple speed limits to cover the various masses of vehicles using the road.

WHAT IF? Suppose a car travels this curve on a wet day and begins to skid on the curve when its speed reaches only 8.00 m/s. What can we say about the coefficient of static friction in this case?

Answer The coefficient of static friction between the tires and a wet road should be smaller than that between the tires and a dry road. This expectation is consistent with experience with driving because a skid is more likely on a wet road than a dry road.

To check our suspicion, we can solve Equation (2) for the coefficient of static friction:

$$\mu_s = \frac{v_{\max}^2}{gr}$$

Substituting the numerical values gives

$$\mu_s = \frac{v_{\max}^2}{gr} = \frac{(8.00\ \text{m/s})^2}{(9.80\ \text{m/s}^2)(35.0\ \text{m})} = 0.187$$

which is indeed smaller than the coefficient of 0.523 for the dry road.

Example 6.4 The Banked Roadway

AM

A civil engineer wishes to redesign the curved roadway in Example 6.3 in such a way that a car will not have to rely on friction to round the curve without skidding. In other words, a car moving at the designated speed can negotiate the curve even when the road is covered with ice. Such a road is usually *banked*, which means that the roadway is tilted toward the inside of the curve as seen in the opening photograph for this chapter. Suppose the designated speed for the road is to be 13.4 m/s (30.0 mi/h) and the radius of the curve is 35.0 m. At what angle should the curve be banked?

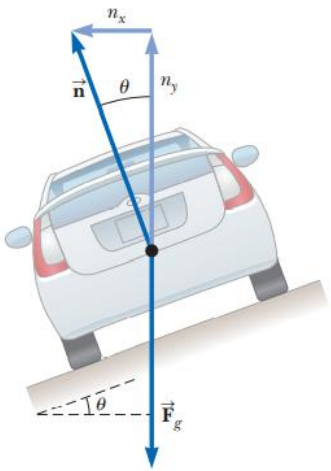


Figure 6.5 (Example 6.4) A car moves into the page and is rounding a curve on a road banked at an angle θ to the horizontal. When friction is neglected, the force that causes the centripetal acceleration and keeps the car moving in its circular path is the horizontal component of the normal force.

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Write Newton's second law for the car in the radial direction, which is the x direction:

$$(1)\quad \sum F_r = n\sin\theta = \frac{mv^2}{r}$$

Apply the particle in equilibrium model to the car in the vertical direction:

$$\sum F_y = n\cos\theta - mg = 0$$

$$(2)\quad n\cos\theta = mg$$

Divide Equation (1) by Equation (2):

$$(3)\quad \tan\theta = \frac{v^2}{rg}$$

Solve for the angle θ:

$$\theta = \tan^{-1}\left[\frac{(13.4\ \text{m/s})^2}{(35.0\ \text{m})(9.80\ \text{m/s}^2)}\right] = 27.6^\circ$$

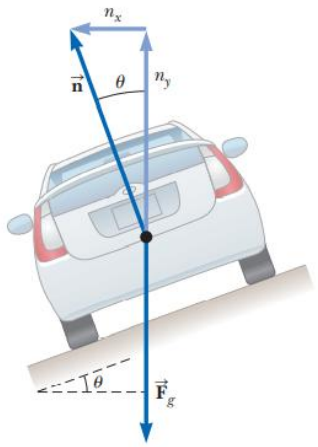


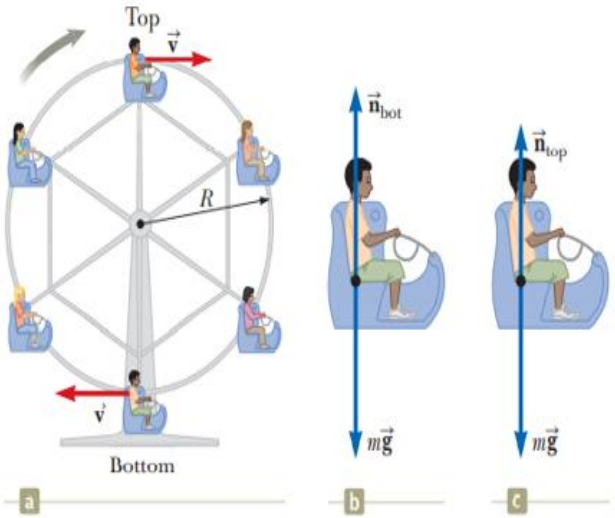
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Example 6.5 Riding the Ferris Wheel

AM

A child of mass m rides on a Ferris wheel as shown in Figure 6.6a. The child moves in a vertical circle of radius 10.0 m at a constant speed of 3.00 m/s.

(A) Determine the force exerted by the seat on the child at the bottom of the ride. Express your answer in terms of the weight of the child, mg.



(B) Determine the force exerted by the seat on the child at the top of the ride.

6.5 continued

Categorize Because the speed of the child is constant, we can categorize this problem as one involving a *particle* (the child) *in uniform circular motion,* complicated by the gravitational force acting at all times on the child.

Analyze We draw a diagram of forces acting on the child at the bottom of the ride as shown in Figure 6.6b. The only forces acting on him are the downward gravitational force $\vec{\mathbf{F}}_g = m\vec{\mathbf{g}}$ and the upward force $\vec{\mathbf{n}}_{\text{bot}}$ exerted by the seat. The net upward force on the child that provides his centripetal acceleration has a magnitude $n_{\text{bot}} - mg$.

Using the particle in uniform circular motion model, apply Newton's second law to the child in the radial direction when he is at the bottom of the ride:

$$\sum F = n_{\text{bot}} - mg = m\frac{v^2}{r}$$

Solve for the force exerted by the seat on the child:

$$n_{\text{bot}} = mg + m\frac{v^2}{r} = mg\left(1 + \frac{v^2}{rg}\right)$$

Substitute numerical values given for the speed and radius:

$$n_{\text{bot}} = mg\left[1 + \frac{(3.00\ \text{m/s})^2}{(10.0\ \text{m})(9.80\ \text{m/s}^2)}\right]$$

$$= 1.09\,mg$$

Hence, the magnitude of the force $\vec{\mathbf{n}}_{\text{bot}}$ exerted by the seat on the child is *greater* than the weight of the child by a factor of 1.09. So, the child experiences an apparent weight that is greater than his true weight by a factor of 1.09.

(B) Determine the force exerted by the seat on the child at the top of the ride.

SOLUTION

Analyze The diagram of forces acting on the child at the top of the ride is shown in Figure 6.6c. The net downward force that provides the centripetal acceleration has a magnitude $mg - n_{top}$.

Apply Newton's second law to the child at this position:
$$\sum F = mg - n_{top} = m\frac{v^2}{r}$$

Solve for the force exerted by the seat on the child:
$$n_{top} = mg - m\frac{v^2}{r} = mg\left(1 - \frac{v^2}{rg}\right)$$

Substitute numerical values:
$$n_{top} = mg\left[1 - \frac{(3.00\ \text{m/s})^2}{(10.0\ \text{m})(9.80\ \text{m/s}^2)}\right]$$
$$= 0.908\,mg$$

In this case, the magnitude of the force exerted by the seat on the child is *less* than his true weight by a factor of 0.908, and the child feels lighter.

Finalize The variations in the normal force are consistent with our prediction in the Conceptualize step of the problem.

WHAT IF? Suppose a defect in the Ferris wheel mechanism causes the speed of the child to increase to 10.0 m/s. What does the child experience at the top of the ride in this case?

Answer If the calculation above is performed with $v = 10.0$ m/s, the magnitude of the normal force at the top of the ride is negative, which is impossible. We interpret it to mean that the required centripetal acceleration of the child is larger than that due to gravity. As a result, the child will lose contact with the seat and will only stay in his circular path if there is a safety bar or a seat belt that provides a downward force on him to keep him in his seat. At the bottom of the ride, the normal force is 2.02 mg, which would be uncomfortable.

44. An object of mass $m_1 = 4.00$ kg is tied to an object of mass $m_2 = 3.00$ kg with String 1 of length $\ell = 0.500$ m. The combination is swung in a vertical circular path on a second string, String 2, of length $\ell = 0.500$ m. During the motion, the two strings are collinear at all times as shown in Figure P6.44. At the top of its motion, m_2 is traveling at $v = 4.00$ m/s. (a) What is the tension in String 1 at this instant? (b) What is the tension in String 2 at this instant? (c) Which string will break first if the combination is rotated faster and faster?

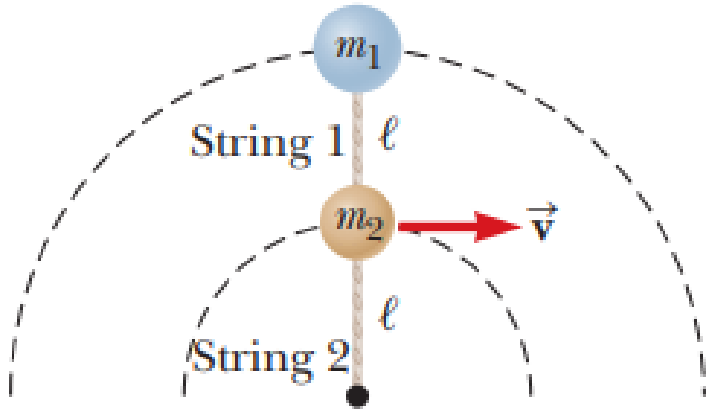
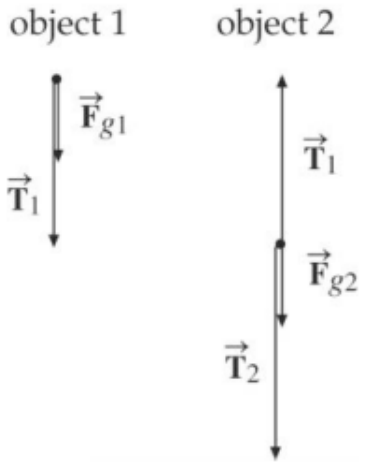


Figure P6.44

P6.44 The radius of the path of object 1 is twice that of object 2. Because the strings are always "collinear," both objects take the same time interval to travel around their respective circles; therefore, the speed of object 1 is twice that of object 2.



S. FIG. P6.44

The free-body diagrams are shown in ANS. FIG. P6.44. We are given $m_1 = 4.00$ kg, $m_2 = 3.00$ kg, $v = 4.00$ m/s, and $\ell = 0.500$ m.

Taking down as the positive direction, we have

$$\text{Object 1: } T_1 + m_1 g = \frac{m_1 v_1^2}{r_1}, \quad \text{where } v_1 = 2v,\ r_1 = 2\ell.$$

$$\text{Object 2: } T_2 - T_1 + m_2 g = \frac{m_2 v_2^2}{r_2}, \quad \text{where } v_2 = v,\ r_2 = 2\ell.$$

(a) From above:

$$T_1 = \frac{m_1 v_1^2}{r_1} - m_1 g = m_1\left(\frac{v_1^2}{r_1} - g\right)$$

$$T_1 = (4.00\text{ kg})\left[\frac{[2(4.00\text{ m/s})]^2}{2(0.500\text{ m})} - 9.80\text{ m/s}^2\right]$$

$$T_1 = 216.8\text{ N} = \boxed{217\text{ N}}$$

(b) From above:

$$T_2 = T_1 + \frac{m_2 v_2^2}{r_2} - m_2 g$$

$$T_2 = T_1 + m_2\left(\frac{v_2{}^2}{r_2} - g\right)$$

$$T_2 = T_1 + (3.00\ \text{kg})\left[\frac{(4.00\ \text{m/s})^2}{0.500\ \text{m}} - 9.80\ \text{m/s}^2\right]$$

$$T_2 = 216.8\ \text{N} + 66.6\ \text{N} = 283.4\ \text{N} = \boxed{283\ \text{N}}$$

(c) $\boxed{\text{From above, } T_2 > T_1 \text{ always, so string 2 will break first.}}$

••49 GO In Fig. 6-39, a car is driven at constant speed over a circular hill and then into a circular valley with the same radius. At the top of the hill, the normal force on the driver from the car seat is 0. The driver's mass is 70.0 kg. What is the magnitude of the normal force on the driver from the seat when the car passes through the bottom of the valley?

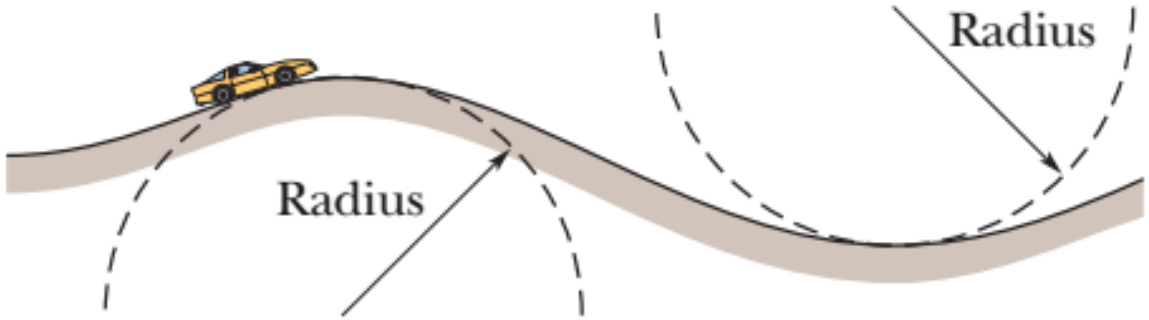


Figure 6-39 Problem 49.

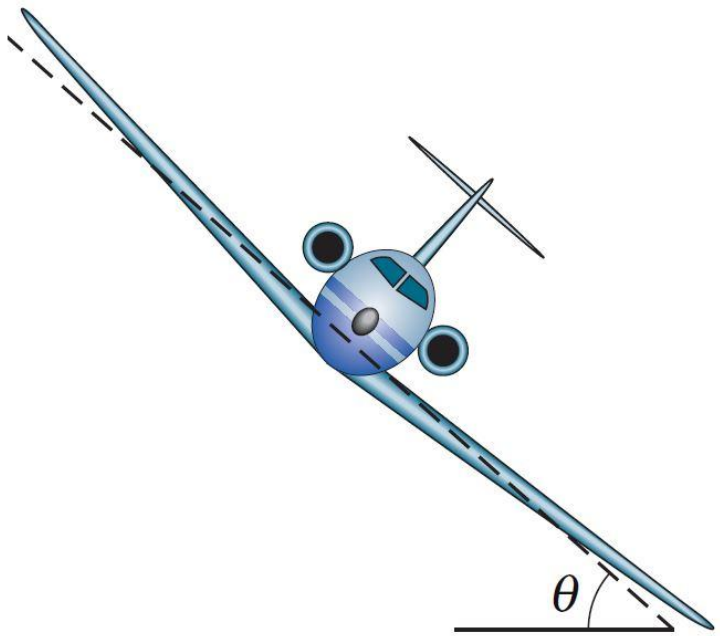
49. At the top of the hill, the situation is similar to that of Sample Problem – "Vertical circular loop, Diavolo," but with the normal force direction reversed. Adapting Eq. 6-19, we find

$$F_N = m(g - v^2/R).$$

Since $F_N = 0$ there (as stated in the problem) then $v^2 = gR$. Later, at the bottom of the valley, we reverse both the normal force direction and the acceleration direction (from what is shown in the Sample Problem) and adapt Eq. 6-19 accordingly. Thus we obtain

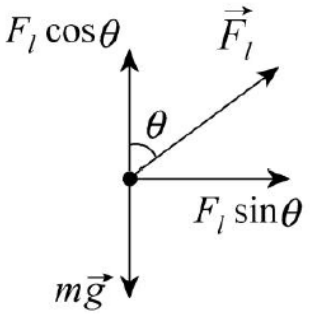
$$F_N = m(g + v^2/R) = 2mg = 1372 \text{ N} \approx 1.37 \times 10^3 \text{ N}.$$

••51 SSM WWW An airplane is flying in a horizontal circle at a speed of 480 km/h (Fig. 6-41). If its wings are tilted at angle $\theta = 40°$ to the horizontal, what is the radius of the circle in which the plane is flying? Assume that the required force is provided entirely by an "aerodynamic lift" that is perpendicular to the wing surface.



51. **THINK** An airplane with its wings tilted at an angle is in a circular motion. Centripetal force is involved in this problem.

EXPRESS The free-body diagram for the airplane of mass m is shown to the right. We note that $\vec{F}_l$ is the force of aerodynamic lift and $\vec{a}$ points rightwards in the figure. We also note that $|\vec{a}| = v^2 / R$. Applying Newton's law to the axes of the problem ($+x$ rightward and $+y$ upward) we obtain



$$F_l \sin\theta = m\frac{v^2}{R}$$
$$F_l \cos\theta = mg$$

Eliminating mass from these equations leads to $\tan\theta = \dfrac{v^2}{gR}$. The equation allows us to solve for the radius R.

ANALYZE With $v = 480$ km/h $= 133$ m/s and $\theta = 40°$, we find

$$R = \frac{v^2}{g\tan\theta} = \frac{(133\ \text{m/s})^2}{(9.8\ \text{m/s}^2)\tan 40°} = 2151\ \text{m} \approx 2.2\times 10^3\ \text{m}.$$

LEARN Our approach to solving this problem is identical to that discussed in the Sample Problem – "Car in banked circular turn." Do you see the similarities?

••57 GO A puck of mass $m = 1.50\,\text{kg}$ slides in a circle of radius $r = 20.0\,\text{cm}$ on a frictionless table while attached to a hanging cylinder of mass $M = 2.50\,\text{kg}$ by means of a cord that extends through a hole in the table (Fig. 6-43). What speed keeps the cylinder at rest?

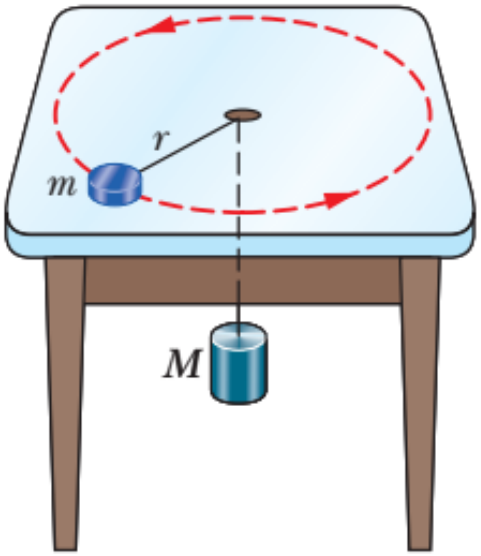
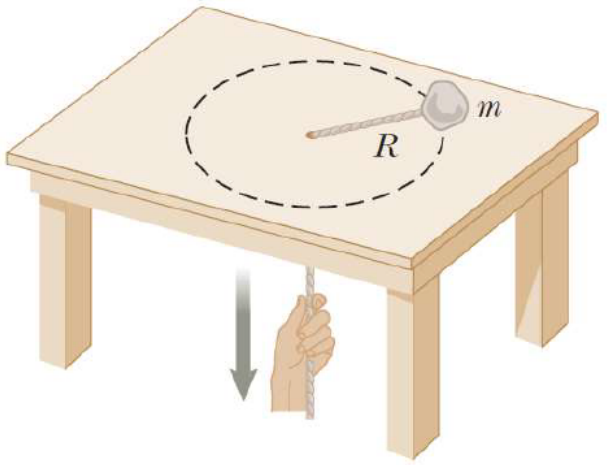


Figure 6-43 Problem 57.

57. For the puck to remain at rest the magnitude of the tension force T of the cord must equal the gravitational force Mg on the cylinder. The tension force supplies the centripetal force that keeps the puck in its circular orbit, so $T = mv^2/r$. Thus $Mg = mv^2/r$. We solve for the speed:

$$v = \sqrt{\frac{Mgr}{m}} = \sqrt{\frac{(2.50\,\text{kg})(9.80\,\text{m/s}^2)(0.200\,\text{m})}{1.50\,\text{kg}}} = 1.81\,\text{m/s}.$$

A string under a tension of 50 N is used to move a rock in a circle of radius 2.5 m at a speed of 20.4 m/s on a frictionless surface as shown in figure. As the string is pulled in, the speed of the rock increases. When the string on the table is 1 m long and the speed of the rock is 51 m/s, the string breaks. Find the maximum tension at which the string break.



33. A 2400-lb (= 10.7-kN) car traveling at 30 mi/h (= 13.4 m/s) attempts to round an unbanked curve with a radius of 200 ft (= 61.0 m). (*a*) What force of friction is required to keep the car on its circular path? (*b*) What minimum coefficient of static friction between the tires and road is required?

34. A circular curve of highway is designed for traffic moving at 60 km/h (= 37 mi/h). (*a*) If the radius of the curve is 150 m (= 490 ft), what is the correct angle of banking of the road? (*b*) If the curve were not banked, what would be the minimum coefficient of friction between tires and road that would keep traffic from skidding at this speed?

35. A conical pendulum is formed by attaching a 53-g pebble to a 1.4-m string. The pebble swings around in a circle of radius 25 cm. (*a*) What is the speed of the pebble? (*b*) What is its acceleration? (*c*) What is the tension in the string?

36. A bicyclist (Fig. 5-41) travels in a circle of radius 25 m at a constant speed of 8.7 m/s. The combined mass of the bicycle and rider is 85 kg. Calculate the force—magnitude and angle with the vertical—exerted by the road on the bicycle.

E5-33 (a) The force required to keep the car in the turn is $F = mv^2/r = Wv^2/(rg)$, or

$$F = (10700\,\text{N})(13.4\,\text{m/s})^2/[(61.0\,\text{m})(9.81\,\text{m/s}^2)] = 3210\,\text{N}.$$

(b) The coefficient of friction required is $\mu_s = F/W = (3210\,\text{N})/(10700\,\text{N}) = 0.300$.

E5-34 (a) The proper banking angle is given by

$$\theta = \arctan\frac{v^2}{Rg} = \arctan\frac{(16.7\,\text{m/s})^2}{(150\,\text{m})(9.8\,\text{m/s}^2)} = 11^\circ.$$

(b) If the road is not banked then the force required to keep the car in the turn is $F = mv^2/r = Wv^2/(Rg)$ and the required coefficient of friction is

$$\mu_s = F/W = \frac{v^2}{Rg} = \frac{(16.7\,\text{m/s})^2}{(150\,\text{m})(9.8\,\text{m/s}^2)} = 0.19.$$

E5-35 (a) This conical pendulum makes an angle $\theta = \arcsin(0.25/1.4) = 10^\circ$ with the vertical. The pebble has a speed of

$$v = \sqrt{Rg\tan\theta} = \sqrt{(0.25\,\text{m})(9.8\,\text{m/s}^2)\tan(10^\circ)} = 0.66\,\text{m/s}.$$

(b) $a = v^2/r = (0.66\,\text{m/s})^2/(0.25\,\text{m}) = 1.7\,\text{m/s}^2$.
(c) $T = mg/cos\theta = (0.053\,\text{kg})(9.8\,\text{m/s}^2)/\cos(10^\circ) = 0.53\,\text{N}$.

E5-36 Ignoring air friction (there must be a forward component to the friction!), we have a normal force upward which is equal to the weight: $N = mg = (85\,\text{kg})(9.8\,\text{m/s}^2) = 833\,\text{N}$. There is a sideways component to the friction which is equal tot eh centripetal force, $F = mv^2/r = (85\,\text{kg})(8.7\,\text{m/s})^2/(25\,\text{m}) = 257\,\text{N}$. The magnitude of the net force of the road on the person is

$$F = \sqrt{(833\,\text{N})^2 + (257\,\text{N})^2} = 870\,\text{N},$$

and the direction is $\theta = \arctan(257/833) = 17^\circ$ off of vertical.

Practice problems

Chapter#6 Problems:

Uniform Circular Motion: 42, 43, 49, 51, 57, 82